

A Unified Four-Nation Five-Currency Equilibrium

SDF Integrability, Reserve-Currency Seigniorage, Cross-Layer No-Arbitrage and
Multidisciplinary Monetary-Geopolitical Equilibrium

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Abstract

This paper develops a unified equilibrium architecture for a world containing at most four nations and at most five simultaneously viable reserve currencies. The national and monetary sides of the system are deliberately separated. National stochastic discount factors (SDFs) form a normalized positive vector on a three-dimensional simplex, while reserve-currency shares form a normalized positive vector on a four-dimensional simplex. The distinction is essential: a four-nation ceiling does not imply a four-currency ceiling.

The paper replaces artificial pairwise conditions such as $F_i F_j = 1$ with a safer no-arbitrage primitive based on relative SDFs, transitivity, integrability, and vanishing closed-cycle products. Bilateral valuation ratios are generated from a common global SDF vector, which automatically eliminates triangular and higher-order arbitrage. A second currency layer is introduced, producing a 4×5 nation-currency valuation matrix. The cross-layer no-arbitrage conditions imply that this matrix is multiplicatively separable and has rank one; equivalently, every 2×2 multiplicative minor vanishes.

The Five-Currency Ceiling is incorporated as a reserve-rent viability constraint: if each viable reserve currency must command at least twenty percent of finite global reserve rents, then five currencies are feasible but six are impossible. At the exact ceiling, all five reserve currencies must have equal shares. In contrast, national SDFs can remain asymmetric. The maximal joint equilibrium is therefore a continuum isomorphic to the interior of a three-dimensional simplex.

A fixed-point theorem establishes existence of at least one joint equilibrium under continuity and positivity. The paper further interprets the architecture using graph theory, gauge-like numeraire invariance, welfare economics, warfare economics, political economy, international relations, evolutionary competition, behavioral economics, physics, chemistry, biology, statistics, computer science, data science, and artificial intelligence. These interdisciplinary interpretations are treated as structural analogies or empirical extensions rather than as substitutes for the core mathematical proof.

The paper ends with “The End”

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1 Introduction

International monetary systems combine several logically distinct objects.

First, nations have stochastic discount factors, valuation kernels, and intertemporal pricing relations. Second, currencies have international reserve demand and settlement functions. Third, reserve-currency issuers may obtain seigniorage rents. Fourth, geopolitical and military capabilities can affect the demand for a currency by influencing the credibility and security of the underlying international system.

A central objective of this paper is to keep these layers distinct while allowing them to interact.

Let

$$N$$

denote the number of nations and let

$$M$$

denote the number of simultaneously viable reserve currencies.

The national result considered here is

$$N \leq 4,$$

whereas the reserve-currency result is

$$M \leq 5.$$

The two bounds concern different objects. The maximal combined architecture is therefore

$$(N, M) = (4, 5).$$

The military wage–seigniorage framework motivating the currency ceiling describes a feedback mechanism

$$\begin{array}{ccccccc} \text{Reserve} & & & & & & \\ \text{Demand} & \rightarrow & \text{Seigniorage} & \rightarrow & \text{Military} & \rightarrow & \text{Military} & \rightarrow & \text{Hegemonic} & \rightarrow & \text{Reserve} \\ & & & & \text{Wages} & & \text{Quality} & & \text{Capability} & & \text{Demand} \end{array}$$

The underlying paper imposes a minimum viable reserve share of twenty percent. Under finite total reserve rents, this yields the Five-Currency Ceiling.

The present paper develops the mathematical architecture surrounding that result.

The principal methodological change is to reject artificial universal conditions such as

$$F_i F_j = 1$$

and instead use the economically meaningful object

$$\frac{F_i}{F_j}.$$

Relative valuation is invariant to common positive rescaling and therefore naturally accommodates numeraire choice.

2 Notation and Basic Objects

Let the nations be indexed by

$$i, j, k, \ell \in \{1, 2, 3, 4\},$$

and reserve currencies by

$$a, b, c, d, e \in \{1, 2, 3, 4, 5\}.$$

Let

$$\mathbf{F} = (F_1, F_2, F_3, F_4)$$

denote the national SDF vector.

Let

$$\mathbf{G} = (G_1, G_2, G_3, G_4, G_5)$$

denote the normalized currency valuation vector.

For national SDFs we impose

$$F_i > 0$$

and

$$\sum_{i=1}^4 F_i = 1.$$

For currencies we impose

$$G_a > 0$$

and

$$\sum_{a=1}^5 G_a = 1.$$

Define the simplices

$$\Delta^3 = \left\{ \mathbf{F} \in \mathbb{R}^4 : F_i \geq 0, \sum_{i=1}^4 F_i = 1 \right\}$$

and

$$\Delta^4 = \left\{ \mathbf{G} \in \mathbb{R}^5 : G_a \geq 0, \sum_{a=1}^5 G_a = 1 \right\}.$$

The interior national simplex is

$$\text{int } \Delta^3 = \{ \mathbf{F} \in \Delta^3 : F_i > 0 \ \forall i \}.$$

Similarly,

$$\text{int } \Delta^4 = \{ \mathbf{G} \in \Delta^4 : G_a > 0 \ \forall a \}.$$

The joint normalized state space is

$$\boxed{\Delta^3 \times \Delta^4}.$$

It has dimension

$$3 + 4 = 7.$$

3 Numeraire Invariance

3.1 Common Rescaling

Suppose that

$$\tilde{F}_i = cF_i$$

for some

$$c > 0.$$

Then

$$\frac{\tilde{F}_i}{\tilde{F}_j} = \frac{cF_i}{cF_j} = \frac{F_i}{F_j}.$$

Thus relative SDFs are invariant under common positive rescaling.

This is the fundamental reason for treating the normalized vector as a representative of an equivalence class.

Define

$$\mathbf{F} \sim \tilde{\mathbf{F}}$$

if there exists $c > 0$ such that

$$\tilde{\mathbf{F}} = c\mathbf{F}.$$

The economically meaningful object is therefore the equivalence class

$$[\mathbf{F}].$$

The normalization

$$\sum_i F_i = 1$$

selects a representative from each positive ray.

3.2 Gauge-Like Interpretation

The transformation

$$\mathbf{F} \mapsto c\mathbf{F}$$

does not alter bilateral ratios. It therefore behaves mathematically like a change of gauge or numeraire.

This analogy is structural rather than a claim that financial markets literally constitute a gauge field in the physical sense.

4 The Safer No-Arbitrage Primitive

Define the bilateral national SDF ratio

$$R_{ij} = \frac{F_i}{F_j}.$$

Then

$$R_{ij}R_{ji} = \frac{F_i}{F_j} \frac{F_j}{F_i} = 1.$$

More importantly,

$$R_{ij}R_{jk} = \frac{F_i}{F_j} \frac{F_j}{F_k} = \frac{F_i}{F_k} = R_{ik}.$$

Therefore

$$R_{ij}R_{jk}R_{ki} = 1.$$

This is the fundamental triangular no-arbitrage identity.

Definition 4.1. *A national SDF system is called globally integrable if there exists a positive vector $\mathbf{F}$ such that every bilateral valuation ratio satisfies*

$$R_{ij} = \frac{F_i}{F_j}.$$

Theorem 4.2 (Closed-Cycle No-Arbitrage). *If bilateral ratios are generated by a common positive SDF vector, then for every closed cycle*

$$i_1 \rightarrow i_2 \rightarrow \cdots \rightarrow i_m \rightarrow i_1,$$

we have

$$\prod_{\ell=1}^m R_{i_\ell i_{\ell+1}} = 1,$$

where $i_{m+1} = i_1$.

Proof. By definition,

$$\prod_{\ell=1}^m R_{i_\ell i_{\ell+1}} = \prod_{\ell=1}^m \frac{F_{i_\ell}}{F_{i_{\ell+1}}}.$$

Every F_i appears once in the numerator and once in the denominator. Hence all terms cancel:

$$\prod_{\ell=1}^m \frac{F_{i_\ell}}{F_{i_{\ell+1}}} = 1.$$

□

5 Differential SDF Integrability

Suppose that SDFs are differentiable along a one-dimensional state variable x .

Then

$$\frac{dF_i}{dF_j} = \frac{F'_i(x)}{F'_j(x)}$$

whenever $F'_j(x) \neq 0$.

Consequently,

$$\frac{dF_i}{dF_j} \frac{dF_j}{dF_k} = \frac{F'_i}{F'_k} = \frac{dF_i}{dF_k}.$$

Therefore,

$$\frac{dF_i}{dF_j} \frac{dF_j}{dF_k} \frac{dF_k}{dF_i} = 1.$$

More generally,

$$\prod_{\ell=1}^m \frac{dF_{i_\ell}}{dF_{i_{\ell+1}}} = 1.$$

The result follows from telescoping cancellation.

In a multidimensional state space, the correct generalization is formulated using differential forms and integrability rather than treating dF_i/dF_j as a globally defined scalar without qualification.

6 The Four-Nation SDF Simplex

With four nations,

$$\mathbf{F} \in \text{int } \Delta^3.$$

The normalized system has three independent degrees of freedom.

A general positive parameterization is

$$F_i = \frac{a_i}{a_1 + a_2 + a_3 + a_4}, \quad a_i > 0.$$

Thus every strictly positive normalized four-nation SDF vector can be represented in this manner. For example, take

$$(a_1, a_2, a_3, a_4) = (2, 1, 3, 4).$$

Then

$$\mathbf{F}^* = \left(\frac{1}{5}, \frac{1}{10}, \frac{3}{10}, \frac{2}{5} \right).$$

The bilateral ratios include

$$R_{12} = 2, \quad R_{23} = \frac{1}{3}, \quad R_{31} = \frac{3}{2}.$$

Hence

$$R_{12}R_{23}R_{31} = 2 \cdot \frac{1}{3} \cdot \frac{3}{2} = 1.$$

Asymmetry therefore does not create arbitrage.

7 The Five-Currency Reserve System

Let

$$x_a = \frac{R_a}{R^W}$$

denote the reserve share of currency a , where

$$R^W = \sum_{a=1}^M R_a$$

is total world reserve demand.

The military wage–seigniorage framework considers seigniorage

$$S_a = sR^W x_a,$$

where s represents an effective reserve-rent or inflation-tax coefficient.

Suppose viability requires

$$S_a \geq S_{\min}.$$

If

$$S_{\min} = \frac{1}{5}sR^W,$$

then

$$sR^W x_a \geq \frac{1}{5}sR^W.$$

Assuming $sR^W > 0$,

$$\boxed{x_a \geq \frac{1}{5}.$$

Since

$$\sum_{a=1}^M x_a = 1,$$

we immediately obtain the Five-Currency Ceiling.

Theorem 7.1 (Five-Currency Ceiling). *If every viable reserve currency must command at least one fifth of finite global reserve rents, then*

$$\boxed{M \leq 5.}$$

Proof. If M reserve currencies are simultaneously viable, then

$$x_a \geq \frac{1}{5}$$

for every a . Hence

$$1 = \sum_{a=1}^M x_a \geq \frac{M}{5}.$$

Therefore

$$M \leq 5.$$

□

8 The Exact Five-Currency Boundary

The ceiling case is stronger than the inequality alone.

Suppose

$$M = 5.$$

Then

$$x_a \geq \frac{1}{5}$$

for every a , while

$$\sum_{a=1}^5 x_a = 1.$$

Define

$$y_a = x_a - \frac{1}{5}.$$

Then

$$y_a \geq 0$$

and

$$\sum_{a=1}^5 y_a = 0.$$

Therefore

$$y_a = 0$$

for every a .

Hence

$$\boxed{x_a = \frac{1}{5} \quad (a = 1, \dots, 5).}$$

Thus five viable currencies can coexist, but at the exact ceiling their reserve shares are necessarily equal.

There is no room for a non-uniform five-currency reserve-share vector under the exact 20% threshold.

9 Why a Sixth Currency Fails

Suppose

$$M = 6.$$

Then viability requires

$$x_a \geq \frac{1}{5}$$

for six currencies.

Consequently,

$$\sum_{a=1}^6 x_a \geq 6 \left(\frac{1}{5} \right) = \frac{6}{5} > 1.$$

But normalization requires

$$\sum_{a=1}^6 x_a = 1.$$

Contradiction.
Thus

$$\boxed{M = 6 \text{ is impossible under the exact model assumptions.}}$$

The result is a direct pigeonhole argument.

10 Currency SDFs

For the joint model, define a normalized currency vector

$$\mathbf{G} = (G_1, G_2, G_3, G_4, G_5).$$

At the five-currency ceiling,

$$\boxed{\mathbf{G}^* = \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5} \right)}.$$

Define bilateral currency valuation ratios

$$Q_{ab} = \frac{G_a}{G_b}.$$

Then

$$Q_{ab}Q_{bc} = Q_{ac}$$

and

$$Q_{a_1 a_2} Q_{a_2 a_3} \cdots Q_{a_m a_1} = 1$$

for every closed currency cycle.

11 The Cross-Layer Valuation Matrix

Define

$$P_{ia} = \frac{F_i}{G_a}.$$

There are

$$4 \times 5 = 20$$

nation-currency links.

The matrix is

$$P = \begin{pmatrix} P_{11} & P_{12} & P_{13} & P_{14} & P_{15} \\ P_{21} & P_{22} & P_{23} & P_{24} & P_{25} \\ P_{31} & P_{32} & P_{33} & P_{34} & P_{35} \\ P_{41} & P_{42} & P_{43} & P_{44} & P_{45} \end{pmatrix}.$$

Holding currency a fixed,

$$\frac{P_{ia}}{P_{ja}} = \frac{F_i/G_a}{F_j/G_a} = \frac{F_i}{F_j}.$$

Holding nation i fixed,

$$\frac{P_{ia}}{P_{ib}} = \frac{F_i/G_a}{F_i/G_b} = \frac{G_b}{G_a}.$$

Thus

$$\boxed{\frac{P_{ia}}{P_{ja}} = R_{ij}}$$

and

$$\boxed{\frac{P_{ia}}{P_{ib}} = Q_{ba}}.$$

12 Rectangular No-Arbitrage

Consider two nations i, j and two currencies a, b .

Then

$$\frac{P_{ia}}{P_{ja}} \frac{P_{jb}}{P_{ib}} = \frac{F_i/F_j}{F_i/F_j} = 1.$$

Hence

$$\boxed{P_{ia}P_{jb} = P_{ib}P_{ja}}.$$

This is the fundamental cross-layer rectangular-cycle condition.

It implies that every 2×2 multiplicative minor vanishes.

Proposition 12.1 (Rank-One Cross-Layer Structure). *If*

$$P_{ia} = \frac{F_i}{G_a},$$

then

$$\boxed{\text{rank}(P) = 1}.$$

Proof. Define

$$H_a = \frac{1}{G_a}.$$

Then

$$P_{ia} = F_i H_a.$$

Hence

$$P = \mathbf{F}\mathbf{H}^\top.$$

It is the outer product of two vectors and therefore has rank one. □

Equivalently,

$$P_{ia}P_{jb} - P_{ib}P_{ja} = 0$$

for all i, j, a, b .

13 Graph-Theoretic Interpretation

The four nations and five currencies form a complete bipartite graph

$$K_{4,5}.$$

It has

$$V = 4 + 5 = 9$$

vertices and

$$E = 4 \times 5 = 20$$

edges.

The dimension of its cycle space is

$$E - V + 1 = 20 - 9 + 1 = 12.$$

Thus the bipartite system has twelve independent cycle constraints.

All other closed cross-layer cycles are generated by these independent cycles.

The graph therefore contains:

- four national nodes;
- five currency nodes;
- twenty nation-currency edges;
- twelve independent cross-layer cycle constraints.

14 A Four-Nation Five-Currency Equilibrium

Let

$$A_i(\mathbf{F}, \mathbf{G}) > 0$$

be continuous national valuation fundamentals and let

$$B_a(\mathbf{F}, \mathbf{G}) > 0$$

be continuous currency fundamentals.

Define

$$T_i^F(\mathbf{F}, \mathbf{G}) = \frac{A_i(\mathbf{F}, \mathbf{G})}{\sum_{j=1}^4 A_j(\mathbf{F}, \mathbf{G})}$$

and

$$T_a^G(\mathbf{F}, \mathbf{G}) = \frac{B_a(\mathbf{F}, \mathbf{G})}{\sum_{b=1}^5 B_b(\mathbf{F}, \mathbf{G})}.$$

Define

$$T(\mathbf{F}, \mathbf{G}) = (T^F(\mathbf{F}, \mathbf{G}), T^G(\mathbf{F}, \mathbf{G})).$$

Then

$$T : \Delta^3 \times \Delta^4 \rightarrow \Delta^3 \times \Delta^4.$$

Theorem 14.1 (Joint Equilibrium Existence). *Suppose that every A_i and B_a is continuous and strictly positive on*

$$\Delta^3 \times \Delta^4.$$

Then there exists at least one

$$(\mathbf{F}^*, \mathbf{G}^*) \in \Delta^3 \times \Delta^4$$

such that

$$\boxed{T(\mathbf{F}^*, \mathbf{G}^*) = (\mathbf{F}^*, \mathbf{G}^*)}.$$

Proof. The product

$$\Delta^3 \times \Delta^4$$

is compact and convex.

Strict positivity of A_i and B_a ensures that their denominators are strictly positive.

Continuity of A_i and B_a therefore implies continuity of T .

By construction,

$$\sum_i T_i^F = 1$$

and

$$\sum_a T_a^G = 1.$$

Thus T maps the product simplex into itself.

Brouwer's fixed-point theorem therefore guarantees a fixed point.

□

15 Explicit Symmetric Equilibrium

Take

$$A_i = 1 \quad (i = 1, \dots, 4)$$

and

$$B_a = 1 \quad (a = 1, \dots, 5).$$

Then

$$F_i^* = \frac{1}{4}$$

and

$$G_a^* = \frac{1}{5}.$$

Therefore

$$\boxed{\mathbf{F}^* = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right)}$$

and

$$\boxed{\mathbf{G}^* = \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5} \right)}.$$

The cross-layer matrix is

$$P_{ia}^* = \frac{1/4}{1/5} = \frac{5}{4}.$$

Hence

$$P^* = \frac{5}{4} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

16 The General Non-Uniform Equilibrium Family

Let

$$a_i > 0$$

and define

$$A = \sum_{i=1}^4 a_i.$$

Set

$$F_i^* = \frac{a_i}{A}.$$

At the five-currency ceiling,

$$G_a^* = \frac{1}{5}.$$

Therefore the general maximal equilibrium family is

$$(\mathbf{F}^*, \mathbf{G}^*) = \left(\frac{a_1}{A}, \frac{a_2}{A}, \frac{a_3}{A}, \frac{a_4}{A}; \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5} \right),$$

where

$$a_i > 0.$$

The equilibrium set is therefore

$$\mathcal{E} = \text{int } \Delta^3 \times \left\{ \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5} \right) \right\}.$$

Consequently,

$$\mathcal{E} \cong \text{int } \Delta^3.$$

The five-currency side is a single point at the exact ceiling, while the national side retains three degrees of freedom.

17 An Explicit Asymmetric Equilibrium

Take

$$(a_1, a_2, a_3, a_4) = (2, 1, 3, 4).$$

Then

$$A = 10$$

and

$$\mathbf{F}^* = \left(\frac{1}{5}, \frac{1}{10}, \frac{3}{10}, \frac{2}{5} \right).$$

The currency vector is

$$\mathbf{G}^* = \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5} \right).$$

The cross-layer matrix becomes

$$P^* = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{3}{2} & \frac{3}{2} & \frac{3}{2} & \frac{3}{2} & \frac{3}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}.$$

Its rank is one.

For example,

$$P_{11}P_{25} = 1 \cdot \frac{1}{2} = \frac{1}{2}$$

and

$$P_{15}P_{21} = 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

Thus

$$P_{11}P_{25} = P_{15}P_{21}.$$

18 Equilibrium Dimensions

The maximal architecture has the following dimensional structure.

Table 1: Dimensional structure of the maximal equilibrium			
Object	Maximum	Normalization	Dimension
Nations	4	$\sum_i F_i = 1$	3
Reserve currencies	5	$\sum_a G_a = 1$	4
Joint system	(4, 5)	Both	7
Five-currency ceiling	5	$x_a \geq 1/5$	0
Maximal equilibrium family	(4, 5)	Both	3

At the exact currency ceiling, the inequality constraints collapse the four-dimensional currency simplex to a single point.

19 Military Wage–Seigniorage Feedback

Let

$$R_t$$

denote reserve-currency demand,

$$S_t$$

seigniorage,

$$M_t$$

military personnel,

$$w_t$$

average military compensation,

$$Q_t$$

military quality,
and

$$H_t$$

hegemonic capability.

The reserve-demand mechanism can be represented as

$$S_t = sR_t^*.$$

Suppose a fraction α of seigniorage finances military wages:

$$B_t = T_t + \alpha S_t.$$

Military labor supply can be represented by

$$M_t^s = L \left(\frac{w_t}{w_t^c} \right),$$

where

$$L' > 0$$

and w_t^c denotes civilian compensation.

Military quality is

$$Q_t = Q(M_t, w_t, A_t),$$

where A_t may represent military artificial intelligence or technological capability.

Hegemonic capability is

$$H_t = H(Q_t, K_t, N_t),$$

where K_t is military capital and N_t is the alliance network.

Reserve demand is then

$$R_t = G(H_t).$$

The feedback loop becomes

$$R_t \rightarrow S_t \rightarrow w_t \rightarrow Q_t \rightarrow H_t \rightarrow R_t.$$

20 Feedback Stability

Define the feedback gain

$$\mathcal{G} = \frac{\partial R}{\partial H} \frac{\partial H}{\partial Q} \frac{\partial Q}{\partial w} \frac{\partial w}{\partial S} \frac{\partial S}{\partial R}.$$

A locally amplifying loop satisfies

$$\boxed{\mathcal{G} > 1.}$$

A locally contracting loop satisfies

$$\boxed{\mathcal{G} < 1.}$$

The boundary case is

$$\mathcal{G} = 1.$$

This framework should be interpreted as a local dynamical-system condition rather than a universal empirical law.

21 Welfare Economics

The joint equilibrium can be embedded in a welfare problem.

Let national welfare be

$$W_i = U_i(C_i, L_i, M_i) + V_i(H_i) - D_i,$$

where C_i is consumption, L_i labor, M_i military labor, H_i security or geopolitical capability, and D_i denotes social costs.

Aggregate welfare is

$$W = \sum_{i=1}^4 \omega_i W_i,$$

where

$$\omega_i > 0$$

are welfare weights.

The existence of a financial equilibrium does not imply that the equilibrium is welfare optimal.

A planner's problem can instead be written as

$$\max_{\mathbf{F}, \mathbf{G}, \mathbf{z}} W(\mathbf{F}, \mathbf{G}, \mathbf{z})$$

subject to

$$\mathbf{F} \in \Delta^3,$$

$$\mathbf{G} \in \Delta^4,$$

and the no-arbitrage and resource constraints.

This distinction separates:

1. competitive equilibrium;
2. monetary stability;
3. geopolitical stability;
4. social welfare optimality.

These need not coincide.

22 Warfare Economics

A military sector can be represented as a specialized labor market.

Let

$$w_i^m$$

denote military compensation and

$$w_i^c$$

civilian compensation.

Define the military wage premium

$$\delta_i = w_i^m - w_i^c.$$

Reserve rents can contribute to this premium:

$$\delta_i = \phi_i S_i, \quad \phi_i \geq 0.$$

The resulting recruitment and retention effects can be represented by

$$M_i = L_i(\delta_i), \quad L_i' > 0.$$

Military human capital can then enter the production function for strategic capability:

$$H_i = H_i(M_i, K_i, A_i).$$

This establishes a labor-allocation channel from monetary rents to strategic capability.

The model does not imply that military expenditure is socially desirable in itself. Military production creates both security benefits and opportunity costs.

23 Political Science and International Relations

Reserve currencies can be interpreted as institutional systems rather than merely monetary objects.

Let credibility be

$$C_i = C(M_i, A_i, I_i),$$

where:

- M_i is military capability;
- A_i is alliance strength;
- I_i is institutional quality.

Reserve demand becomes

$$R_i = R(C_i, F_i, \mathcal{T}_i),$$

where $\mathcal{T}_i$ represents trade-network centrality.

The resulting system has a strategic interaction structure:

Currency credibility $\leftrightarrow$ Alliance credibility $\leftrightarrow$ Military capability $\leftrightarrow$ Reserve demand.

International monetary equilibrium is consequently connected to international political equilibrium.

24 Diplomacy and Network Centrality

Let

$$A = (A_{ij})$$

be an alliance adjacency matrix.

Let

$$\mathbf{c}$$

be a centrality vector.

A simplified centrality equation is

$$\mathbf{c} = \lambda A \mathbf{c}.$$

Thus centrality corresponds to an eigenvector problem.

Reserve-currency demand can depend upon centrality:

$$R_i = R_i(c_i, H_i, F_i).$$

A currency issuer with greater network centrality may have a larger potential market for its currency, conditional on institutional and economic credibility.

25 Evolutionary Interpretation

Suppose reserve currencies compete for international adoption.

Let

$$x_a(t)$$

denote currency a 's share.

A replicator-style dynamic is

$$\dot{x}_a = x_a (\pi_a - \bar{\pi}),$$

where

$$\bar{\pi} = \sum_b x_b \pi_b.$$

Here π_a represents an aggregate currency fitness function involving financial depth, credibility, geopolitical security, network effects, and reserve rents.

The exact five-currency ceiling corresponds to a constrained boundary in which every surviving currency receives exactly the minimum viable share.

This is an evolutionary analogy: currencies that cannot maintain the required international rent share become nonviable.

26 Behavioral Economics and Psychology

International reserve demand is not necessarily determined solely by static optimization.

Perceived safety, institutional trust, salience, familiarity, and expectations can influence reserve demand.

Let

$$\theta_a$$

represent perceived credibility and

$$\chi_a$$

represent confidence.

A behavioral reserve-demand function can be written

$$R_a = R_a(F_a, C_a, \theta_a, \chi_a, \mathcal{N}_a),$$

where $\mathcal{N}_a$ captures network effects.

A simple bounded-confidence specification is

$$\theta_a = \sigma(\beta_0 + \beta_1 C_a + \beta_2 \chi_a),$$

where

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

This creates a channel through which expectations can amplify or attenuate the monetary-geopolitical feedback loop.

The psychological interpretation is deliberately restricted to aggregate decision-making and expectations; it does not assert clinical or psychiatric properties of individuals.

27 Physics Analogy

The national ratios

$$R_{ij} = \frac{F_i}{F_j}$$

have a multiplicative path structure.

Taking logarithms gives

$$r_{ij} = \log R_{ij} = \log F_i - \log F_j.$$

For a closed cycle,

$$\sum_{\ell=1}^m r_{i_\ell i_{\ell+1}} = 0.$$

Thus the logarithmic valuation field is conservative.

This resembles a zero-holonomy condition in differential geometry.

The analogy is useful because it emphasizes the distinction between:

- local bilateral valuation;
- global path independence;
- closed-cycle consistency.

It should not be interpreted as claiming that financial SDFs literally obey Maxwell equations or physical gauge dynamics.

28 Chemistry Analogy

The multiplicative cycle condition can also be represented using reaction-network language.

Suppose each bilateral exchange is represented by a reaction-like transformation with factor

$$R_{ij}.$$

A closed reaction cycle has equilibrium product

$$R_{12}R_{23}R_{31} = 1.$$

Taking logarithms,

$$\log R_{12} + \log R_{23} + \log R_{31} = 0.$$

This resembles the thermodynamic requirement that a reversible equilibrium network have zero net affinity around a closed equilibrium cycle.

Again, the correspondence is structural rather than literal.

29 Computer Science Representation

The equilibrium system can be represented as a graph database.

Each nation is a node

$$N_i$$

and each currency is a node

$$C_a.$$

Each bilateral valuation is an edge.

The nation-currency graph contains

$$20$$

edges.

The rank-one condition means that the edge weights are generated by node potentials:

$$P_{ia} = F_i G_a^{-1}.$$

Hence edge information is not independent.

This reduces the effective representation from twenty arbitrary edge weights to the normalized node variables

$$(F_1, F_2, F_3, F_4, G_1, \dots, G_5).$$

After normalization, this is seven degrees of freedom.

30 Algorithmic Equilibrium Computation

A practical numerical algorithm can iterate the normalized maps.

Pseudo-Code

INPUT:

Initial $F[1:4] > 0$

Initial $G[1:5] > 0$

Continuous positive functions $A_i(F, G)$

Continuous positive functions $B_a(F, G)$

Tolerance ϵ

Maximum iterations K

NORMALIZE F :

$F[i] = F[i] / \sum_j F[j]$

NORMALIZE G :

$G[a] = G[a] / \sum_b G[b]$

FOR $k = 1, \dots, K$:

FOR $i = 1, \dots, 4$:

$F_{\text{new}}[i] = A_i(F, G)$

$F_{\text{new}} = F_{\text{new}} / \sum_j F_{\text{new}}[j]$

FOR $a = 1, \dots, 5$:

$G_{\text{new}}[a] = B_a(F, G)$

$G_{\text{new}} = G_{\text{new}} / \sum_b G_{\text{new}}[b]$

```

IF norm(F_new-F) + norm(G_new-G) < epsilon:
    RETURN F_new, G_new

```

```

F = F_new
G = G_new

```

```

RETURN last iterate

```

Convergence of this particular iteration is not guaranteed by Brouwer alone. Brouwer establishes existence, not convergence of an arbitrary algorithm.

If the map is a contraction, Banach's fixed-point theorem can instead provide uniqueness and convergence.

31 Statistical Identification

The theoretical system suggests several empirical objects.

A reserve-share equation can be written as

$$x_{it} = \alpha_i + \beta_1 M_{it} + \beta_2 W_{it} + \beta_3 A_{it} + \beta_4 F_{it} + \varepsilon_{it},$$

where:

- M_{it} is military expenditure or capability;
- W_{it} is the military wage bill;
- A_{it} is alliance centrality;
- F_{it} is financial depth.

The military wage–seigniorage mechanism suggests the hypothesis

$$\beta_2 > 0,$$

conditional on appropriate identification assumptions.

A dynamic specification may include lagged reserve shares:

$$x_{it} = \rho x_{i,t-1} + \beta' Z_{it} + \varepsilon_{it}.$$

A panel design can incorporate issuer fixed effects and time fixed effects:

$$x_{it} = \alpha_i + \gamma_t + \beta' Z_{it} + \varepsilon_{it}.$$

The theory does not itself identify causal coefficients; empirical identification requires exogenous variation or an appropriate quasi-experimental design.

32 Data Science and Machine Learning

The system naturally generates a constrained machine-learning problem.

Given observations

$$\{(\mathbf{F}_t, \mathbf{G}_t, \mathbf{Z}_t)\}_{t=1}^T,$$

one may estimate

$$\hat{T}_\theta(\mathbf{F}_t, \mathbf{G}_t, \mathbf{Z}_t)$$

using a parametric or neural model.

The output can be constrained by a softmax transformation:

$$F_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

and

$$G_a = \frac{e^{y_a}}{\sum_b e^{y_b}}.$$

This guarantees positivity and normalization.

For the currency layer, a ceiling-aware model can impose

$$G_a = \frac{1}{5} + \left(1 - \frac{5}{5}\right) u_a,$$

only when the minimum-share constraint is relaxed. Under the exact five-currency ceiling, the model instead collapses to

$$G_a = \frac{1}{5}.$$

The rank-one condition can be incorporated into a machine-learning loss:

$$\mathcal{L} = \mathcal{L}_{\text{prediction}} + \lambda \sum_{i,j,a,b} (P_{ia}P_{jb} - P_{ib}P_{ja})^2.$$

This is a physics-informed or economically constrained learning architecture.

33 Artificial Intelligence Extension

A world model can represent the state as

$$s_t = (\mathbf{F}_t, \mathbf{G}_t, \mathbf{R}_t, \mathbf{S}_t, \mathbf{M}_t, \mathbf{H}_t, \mathbf{A}_t).$$

A learned transition model is

$$s_{t+1} = \mathcal{W}_\theta(s_t, a_t, \epsilon_t).$$

A policy can then choose actions affecting:

- fiscal expenditure;
- military wages;
- financial regulation;
- reserve-market intervention;
- diplomatic investment;
- alliance formation;
- technological investment.

The objective can be written as

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \beta^t U(s_t, a_t) \right].$$

The equilibrium constraints remain hard structural restrictions even when the transition model is learned.

34 Multi-Agent Interpretation

Let each nation be an agent.

Nation i chooses an action vector

$$a_i = (f_i, m_i, d_i, k_i),$$

representing fiscal policy, military investment, diplomacy, and capital investment.

The joint action is

$$\mathbf{a} = (a_1, a_2, a_3, a_4).$$

The state includes currency shares

G.

A Markov game can then be written as

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}_1, \dots, \mathcal{A}_4, P, U_1, \dots, U_4).$$

The reserve-currency system becomes an endogenous component of the strategic game.

A Nash equilibrium satisfies

$$U_i(a_i^*, a_{-i}^*) \geq U_i(a_i, a_{-i}^*)$$

for every feasible unilateral deviation.

This strategic equilibrium is distinct from the competitive SDF fixed point, although the two can be embedded in a unified model.

35 Physical Security and Reserve Demand

Let

$$C_a$$

represent currency credibility.

Suppose

$$C_a = C_a(F_a, H_a, I_a, A_a, T_a),$$

where the arguments represent financial strength, hegemonic capability, institutions, alliance networks, and trade centrality.

Reserve demand becomes

$$R_a = R_a(C_a).$$

Seigniorage is

$$S_a = sR_a.$$

Military financing is

$$B_a = T_a + \alpha_a S_a.$$

Thus

$$R_a \rightarrow S_a \rightarrow B_a \rightarrow H_a \rightarrow C_a \rightarrow R_a.$$

This provides a formal channel through which monetary privilege can become strategic capacity.

36 Comparative Statics

At the exact five-currency ceiling,

$$x_a = \frac{1}{5}.$$

An increase in the minimum viability threshold from $1/5$ to $\underline{x}$ changes the maximum number of currencies to

$$M_{\max} \leq \left\lfloor \frac{1}{\underline{x}} \right\rfloor.$$

Thus

$$M_{\max} = \left\lfloor \frac{1}{\underline{x}} \right\rfloor$$

under the simplified finite-rent model.

For example,

$$\underline{x} = 0.20 \quad \Rightarrow \quad M_{\max} = 5.$$

If

$$\underline{x} = 0.25,$$

then

$$M_{\max} = 4.$$

If

$$\underline{x} = 0.10,$$

then

$$M_{\max} = 10.$$

The ceiling is therefore endogenous to the viability threshold.

37 A Unified Equilibrium Diagram

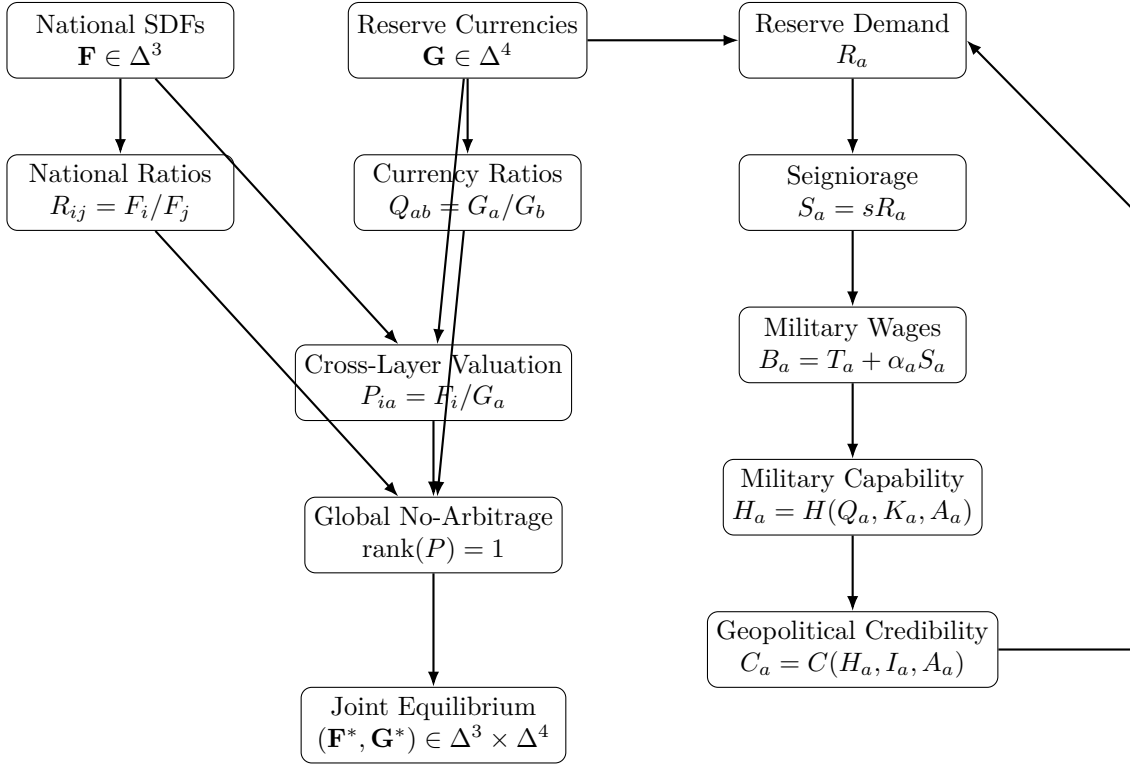


Figure 1: A unified equilibrium diagram.

The diagram separates the mathematical no-arbitrage structure from the military-geopolitical feedback mechanism while allowing the currency state to connect the two.

38 Physical, Chemical, Biological, and Computational Analogies

The mathematical architecture has several interdisciplinary interpretations.

38.1 Physics

The logarithmic bilateral valuation

$$r_{ij} = \log F_i - \log F_j$$

is a potential difference. Closed-loop sums vanish.

38.2 Chemistry

The same algebra describes a reaction network in which equilibrium affinities cancel around closed reversible cycles.

38.3 Biology

Reserve currencies can be interpreted as competing institutional “strategies” whose population shares evolve through relative fitness. Network effects create frequency dependence.

38.4 Psychology

Expectations and perceived safety can shift reserve demand even before fundamental variables change.

38.5 Political Science

Reserve currency status represents institutional and strategic credibility.

38.6 International Relations

Currency networks and alliance networks can reinforce one another.

38.7 Warfare Economics

Seigniorage rents can finance specialized military labor and technology, which can feed back into geopolitical credibility.

38.8 Computer Science

The equilibrium is a constrained fixed-point problem on a product simplex.

38.9 Artificial Intelligence

The system can be embedded in a learned world model or multi-agent reinforcement-learning environment, subject to exact normalization and no-arbitrage constraints.

These interpretations are complementary. None should be interpreted as evidence that the mathematical system is literally identical to the corresponding physical, chemical, biological, psychological, or computational system.

39 The Main Structural Theorem

We can now combine the principal results.

Theorem 39.1 (Four-Nation Five-Currency Joint Equilibrium). *Suppose:*

1. *the number of nations satisfies*

$$N \leq 4;$$

2. *reserve currencies face a finite total reserve-rent constraint;*

3. *every viable reserve currency requires at least*

$$\frac{1}{5}$$

of global reserve rents;

4. *national SDF fundamentals are continuous and strictly positive;*

5. *currency fundamentals are continuous and strictly positive;*

6. *bilateral national and currency valuations are generated by common normalized SDF vectors.*

Then:

1. *the number of viable reserve currencies satisfies*

$$M \leq 5;$$

2. *if $M = 5$, then*

$$G_a = \frac{1}{5}$$

for every a ;

3. *the national SDF vector may be any element of*

$$\text{int } \Delta^3;$$

4. the cross-layer valuation matrix

$$P_{ia} = \frac{F_i}{G_a}$$

has rank one;

5. every closed national, currency, and rectangular cross-layer arbitrage cycle has unit product;

6. at least one joint equilibrium exists.

Proof. Part (1) follows from the Five-Currency Ceiling theorem.

Part (2) follows because five numbers each bounded below by $1/5$ must sum to one and therefore must all equal $1/5$.

Part (3) follows from positivity and national normalization.

Part (4) follows from

$$P_{ia} = F_i G_a^{-1},$$

which is an outer product.

Part (5) follows from telescoping cancellation.

Part (6) follows from Brouwer's fixed-point theorem applied to the continuous map

$$T : \Delta^3 \times \Delta^4 \rightarrow \Delta^3 \times \Delta^4.$$

Therefore all six statements hold. □

40 Economic Interpretation of the Maximal State

The maximal configuration is

$$\boxed{N = 4, \quad M = 5.}$$

It should not be interpreted as four currencies.

The system contains four national SDF nodes and five reserve-currency nodes.

The maximal cross-layer network therefore has

$$4 + 5 = 9$$

nodes and

$$4 \times 5 = 20$$

nation-currency links.

The five-currency boundary has

$$5 \times 20\% = 100\%$$

of the available reserve-rent allocation.

The national SDF system, however, retains three degrees of freedom.

Thus monetary scarcity and national valuation flexibility coexist.

41 Empirical Research Program

The theory suggests a multi-layer empirical program.

41.1 Layer I: National SDF Estimation

Estimate

$$F_{it}$$

using asset-pricing moments:

$$E_t[m_{t+1}R_{i,t+1}] = 1.$$

Then test whether bilateral ratios satisfy

$$R_{ij,t} \approx \frac{F_{it}}{F_{jt}}.$$

41.2 Layer II: Currency Reserve Shares

Estimate

$$x_{at} = \frac{R_{at}}{R_t^W}.$$

Test whether reserve shares exhibit persistent concentration and whether the observed distribution approaches the theoretical threshold.

41.3 Layer III: Military Wage Channel

Estimate

$$S_{at} = s_a R_t^W x_{at}$$

and test

$$\frac{\partial w_{at}}{\partial S_{at}} > 0.$$

41.4 Layer IV: Geopolitical Feedback

Estimate

$$H_{at} = \alpha + \beta_1 Q_{at} + \beta_2 K_{at} + \beta_3 A_{at} + \varepsilon_{at}.$$

Then estimate reserve demand as a function of geopolitical capability.

41.5 Layer V: Cross-Layer Rank Test

Construct an empirical valuation matrix

$$\hat{P}_{iat}.$$

The theoretical restriction is

$$\text{rank}(P_t) = 1.$$

An empirical test can compare the first singular value to the remaining singular values.

Let

$$\sigma_1 \geq \sigma_2 \geq \dots$$

be the singular values of $\hat{P}_t$.

The theory predicts

$$\boxed{\sigma_2 = \sigma_3 = \dots = 0}$$

in the exact model.

In real data one expects sampling error, measurement error, institutional frictions, and model misspecification to produce nonzero residual singular values.

42 Robustness and Limitations

Several qualifications are important.

First, the Five-Currency Ceiling depends on the assumed twenty-percent viability threshold.

Second, the existence theorem requires continuity and positivity. Brouwer does not provide uniqueness.

Third, the rank-one cross-layer condition is exact only if all bilateral valuations are generated from common scalar national and currency SDF vectors.

Fourth, actual financial markets contain transaction costs, bid-ask spreads, capital controls, taxes, default risk, settlement frictions, and heterogeneous funding constraints.

Fifth, geopolitical and military variables are endogenous. Correlation between military power and reserve-currency status does not establish the direction of causality.

Sixth, the interdisciplinary analogies are conceptual. They do not constitute empirical validation.

43 Conclusion

The economically cleaner formulation replaces artificial pairwise conditions with a global normalized SDF architecture.

The national system is

$$\mathbf{F} \in \text{int } \Delta^3,$$

while the reserve-currency system is

$$\mathbf{G} \in \Delta^4.$$

The two ceilings are distinct:

$$\boxed{N_{\max} \leq 4, \quad M_{\max} \leq 5.}$$

At the exact five-currency boundary,

$$\boxed{G_1 = G_2 = G_3 = G_4 = G_5 = \frac{1}{5}.}$$

The national SDFs can nevertheless be asymmetric.

The cross-layer valuation matrix

$$P_{ia} = \frac{F_i}{G_a}$$

has rank one:

$$\boxed{\text{rank}(P) = 1.}$$

Consequently,

$$P_{ia}P_{jb} = P_{ib}P_{ja}$$

for every pair of nations and currencies.

All closed valuation cycles therefore have unit product.

The maximal equilibrium family is

$$\boxed{\mathcal{E} = \text{int } \Delta^3 \times \left\{ \left(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5} \right) \right\}.}$$

The exact five-currency ceiling is therefore not an obstacle to equilibrium. It is a boundary equilibrium in which reserve-rent capacity is completely allocated among five currencies, while national SDF heterogeneity remains possible.

The resulting architecture is

$$\boxed{4 \text{ nations} \times 5 \text{ reserve currencies} = 20 \text{ nation-currency links},}$$

with

$$\boxed{9 \text{ nodes}}$$

and

$$\boxed{12 \text{ independent cross-layer cycles.}}$$

The central mathematical principle is simple:

$$\boxed{\text{Global consistency is stronger and cleaner than arbitrary pairwise restrictions.}}$$

The national and currency layers are normalized separately, their relative valuations are generated from common potentials, their closed cycles have zero logarithmic holonomy, and their joint equilibrium exists under standard continuity and positivity conditions.

44 Appendix A: Proof of the Cycle-Space Dimension

For a connected graph with V vertices and E edges, the dimension of its cycle space is

$$E - V + 1.$$

For the complete bipartite graph $K_{4,5}$,

$$V = 9$$

and

$$E = 20.$$

Therefore

$$\dim(\mathcal{C}) = 20 - 9 + 1 = 12.$$

Hence twelve independent cycles generate the complete cycle space.

45 Appendix B: Proof of the Rank-One Representation

Suppose

$$P_{ia} = \frac{F_i}{G_a}.$$

Define

$$\mathbf{u} = \begin{pmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{pmatrix}$$

and

$$\mathbf{v} = \begin{pmatrix} 1/G_1 \\ 1/G_2 \\ 1/G_3 \\ 1/G_4 \\ 1/G_5 \end{pmatrix}.$$

Then

$$P = \mathbf{u}\mathbf{v}^\top.$$

Therefore

$$\text{rank}(P) \leq 1.$$

Since all F_i and G_a are positive,

$$P \neq 0,$$

so

$$\boxed{\text{rank}(P) = 1.}$$

46 Appendix C: Alternative Logarithmic Representation

Define

$$f_i = \log F_i, \quad g_a = \log G_a.$$

Then

$$\log R_{ij} = f_i - f_j$$

and

$$\log Q_{ab} = g_a - g_b.$$

The cross-layer valuation satisfies

$$\log P_{ia} = f_i - g_a.$$

Thus every bilateral valuation is a difference of node potentials.
For a closed cycle,

$$\sum_{\ell} (f_{i_\ell} - f_{i_{\ell+1}}) = 0.$$

The same applies to currencies and cross-layer cycles.
This is the additive representation of global no-arbitrage.

47 Appendix D: Sensitivity of the Currency Ceiling

Let the minimum viable reserve share be

$$\underline{x}.$$

Then

$$x_a \geq \underline{x}$$

implies

$$1 = \sum_{a=1}^M x_a \geq M \underline{x}.$$

Therefore

$$M \leq \frac{1}{\underline{x}}.$$

Since M is an integer,

$$\boxed{M_{\max} = \left\lfloor \frac{1}{\underline{x}} \right\rfloor.}$$

The Five-Currency Ceiling is therefore the special case

$$\underline{x} = \frac{1}{5}.$$

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Glossary

SDF

Stochastic discount factor; an intertemporal valuation kernel used to price uncertain payoffs.

National SDF

The valuation function F_i associated with nation i .

Currency share

The fraction x_a of global reserve demand allocated to currency a .

Reserve currency

A currency held internationally as a reserve, settlement, or store-of-value asset.

Seigniorage

The economic rent associated with issuing money or providing internationally demanded monetary liabilities.

Military wage seigniorage

The mechanism through which monetary rents can contribute to military compensation.

Reserve-rent viability threshold

The minimum reserve share required for a currency to finance the assumed minimum strategic capability.

No-arbitrage

The condition that a costless closed trading strategy cannot generate a positive risk-adjusted gain.

Integrability

The property that bilateral valuations can be represented by differences or ratios of global potentials.

Holonomy

The accumulated transformation around a closed path; in the present model it is unity multiplicatively and zero logarithmically.

Numeraire

The unit in which relative prices are expressed.

Simplex

The set of nonnegative vectors whose components sum to one.

Product simplex

The Cartesian product $\Delta^3 \times \Delta^4$ describing the joint normalized national-currency state.

Cross-layer valuation

The valuation $P_{ia} = F_i/G_a$ linking national and currency layers.

Rank-one structure

The property that the cross-layer valuation matrix can be written as an outer product of two vectors.

Rectangular cycle

A four-edge cycle involving two nations and two currencies.

Cycle space

The vector space generated by graph cycles; its dimension is $E - V + 1$ for a connected graph.

Feedback gain

The product of marginal derivatives around a dynamic feedback loop.

Military capability

Aggregate strategic capacity generated by military labor, capital, technology, and organization.

Alliance centrality

A network measure of the strategic position of an actor within an alliance system.

Fixed point

A state satisfying $T(x) = x$.

Brouwer equilibrium existence

The guarantee that every continuous self-map of a nonempty compact convex subset of Euclidean space has a fixed point.

Welfare equilibrium

An equilibrium evaluated according to a social objective rather than merely feasibility or market clearing.

Replicator dynamics

Evolutionary dynamics in which the growth of a strategy's share depends on its payoff relative to average payoff.

World model

A learned dynamical representation of an environment used for prediction and planning.

Multi-agent reinforcement learning

Reinforcement learning in which multiple strategic agents interact in a common environment.

Rank constraint

A restriction on the dimensionality of a matrix, here expressing multiplicative separability of nation-currency valuations.

The End